

Building the AgiMo Digital Twin: First steps towards continuous transport model assimilation

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Abstract

Agent-based transport models typically rely on static base-year calibrations, limiting their ability to reflect day-to-day variations in travel behavior driven by weather, public transport reorganizations, or calendar events. This paper presents the ongoing research towards a regular updated digital twin for the Berlin-Brandenburg region built upon the MATSim Open Berlin scenario. Utilizing mobile-phone-derived out-of-home data to update the synthetic population, the model assimilates day-specific. Evaluated across 12 diverse Tuesdays in 2023, the assimilated model improves alignment with real-world traffic counts slightly with a 1% increase of the avg SQV, while keeping the mode choice stable compared to the baseline references. The runtimes show that overnight processing is feasible. This setup provides a synchronized, day-specific baseline for policy evaluation, paving the way for testing new algorithms and integrating floating car data, weather impacts, and dynamic transit schedules.

1. Introduction & Motivation

Agent-based transport models are typically calibrated once, against a base year, and then used for scenario analysis over several years. Calibration is labour-intensive and largely manual, so the base year tends to age: land use changes, public transport supply is reorganised, and travel behaviour shifts – as the post-pandemic period has demonstrated at an unusual pace. Further the typical approach does not consider the changes in travel behavior between certain weekdays or between different weather conditions. The Open Berlin scenario follows this pattern: it reproduces mean weekday conditions well, but it has no mechanism to track how the real system behaved on a *given* day.

The established modelling approach is robust and well investigated, which is why traditional household travel surveys such as the SrV (Hubrich et al., 2019) remain the methodological backbone of transport modeling. In recent years, however, new data sources such as **mobile phone data** have emerged. Although research on these sources is still at an early stage, they promise new opportunities for developing transport models that reflect reality more closely (Wang et al., 2018).

The concept of a regularly updated digital copy of a physical system is also known as **digital twin**. First developed in the domain of Product Lifecycle Management (Grieves, 2015) the concept is coming more and more to other domains like smart cities as Fuller et al., 2020 state. Although there are different approaches to create digital twins of transport systems in a standardized way (e.g. Grübel et al., 2023), we are currently not aware of any attempt to create a system-wide digital twin that is updated on a regular basis.

We report on our ongoing effort within the CRC AgiMo¹ to operate the *Open Berlin Scenario* (Rakow et al., 2026) as a **digital twin** build on MATSim (Horni et al., 2016): a simulation that is re-run for specific days, automatically assimilated using mobile-phone-derived out-of-home data, and continuously validated against traffic-count-data. The aim is not a real-time control system, but a model that (i) stays synchronised with the observed state of the network and

¹<https://gepris.dfg.de/project/565177237>, last access 2026-08-14

(ii) produces a validated, day-specific baseline for policy questions. We will present first results of our semi-automated pipeline and give an outlook to our future research.

2. Data & Methodology

Mobile phone data. Senozon (Senozon Deutschland GmbH, 2026) supplies mobile-phone-based out-of-home-data on zip-code-level. The data is day-specific, but they carry known biases: low cell-tower-resolution in rural areas, under-detection of short trips and no observed mode. We therefore use the data as a baseline for the *spatial change* of demand rather than for absolute values.

Open Berlin Scenario. We use the MATSim Open Berlin Scenario v7.1.1 (1%) which covers Berlin and the surrounding districts. For the 100%-scenario it contains approximately 6 million agents. We use this scenario as is with the existing parameter calibration.

Traffic counts. We use approximately 280 permanent counting stations in the Berlin/Brandenburg area from the *City of Berlin*², delivered as hourly volumes per direction.

Digital twin setup. The setup utilizes the Open Berlin scenario and consists of six stages, orchestrated as a automated pipeline. As input, mobile-phone-based out-of-home-data is converted to a unified index (0) and temporally aggregated (1). It is then normalised against a pre-defined baseline (2) and used to assimilate the synthetic MATSim population by modifying the zipcode-specific out-of-home-rate according to the generated baseline (3). This is resulting in an adapted MATSim-Population that is finally simulated (4) and compared against permanent counting stations (5).

For the initial setup we defined 12 different Tuesdays in 2023 as test data set as shown in Figure 1. The days were selected to reflect varying weather conditions, as well as school holidays and public holidays. This allows us to evaluate future improvements of any new implemented methodology for assimilation.

3. Preliminary results

We will present our explorative data-analysis on the described input-data to show the relevance of our approach. Further we will show our preliminary results of the execution of our pipeline. We were able to determine the following:

- **Fit to counts.** We compared the daily delivered hourly count data with the average weekday-model and with the assimilated daily model. Over all stations, timebins and days the average SQV increased by 1%. Figure 1 shows this on a more detailed, daily level.
- **Behavioural plausibility.** Mode shares remain within 1 percentage point of the average weekday reference over the full period, indicating that daily re-anchoring does not progressively distort choice behaviour.
- **Anomaly days.** October 3rd demonstrates that days differing significantly from the average workday require a different approach.
- **Cost.** One simulated day requires 12 hours on a node of TUB HPC³ with 12 threads for the QSim and 16 threads for replanning, i.e. the processing is possible within an overnight time window.

4. Discussion and outlook

Demonstrating the feasibility of daily updates for a 1% transport model covering the 6 million inhabitants of Berlin-Brandenburg provides a solid baseline, though the underlying algorithms

²<https://api.viz.berlin.de/daten/verkehrsdetektion>, last access: 2026-07-31

³<https://www.tu.berlin/campusmanagement/angebot/high-performance-computing-hpc>, last access 2026-08-

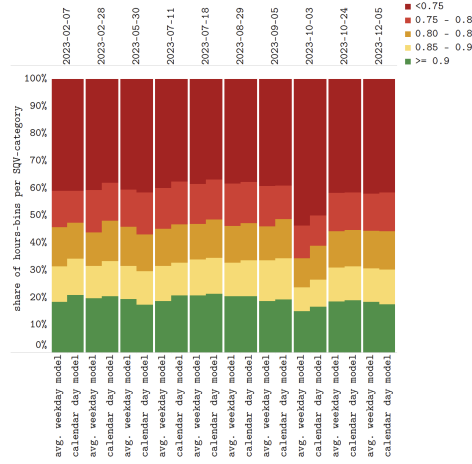


Figure 1: Hourly count-data of a specific day compared with the simulated values of the avg. weekday model and the calendar day model from the digital twin for this day using the categorized hourly SQV.

require further methodological refinement. Key upcoming enhancements include integrating mobile phone activity-time histograms, broadening validation with bike-count data, and evaluating Floating Car Data (FCD) for the region. Additionally, future iterations will assess the impact of weather conditions and incorporate publicly available incident reports, such as road closures and accidents and also day-specific transit schedules.

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